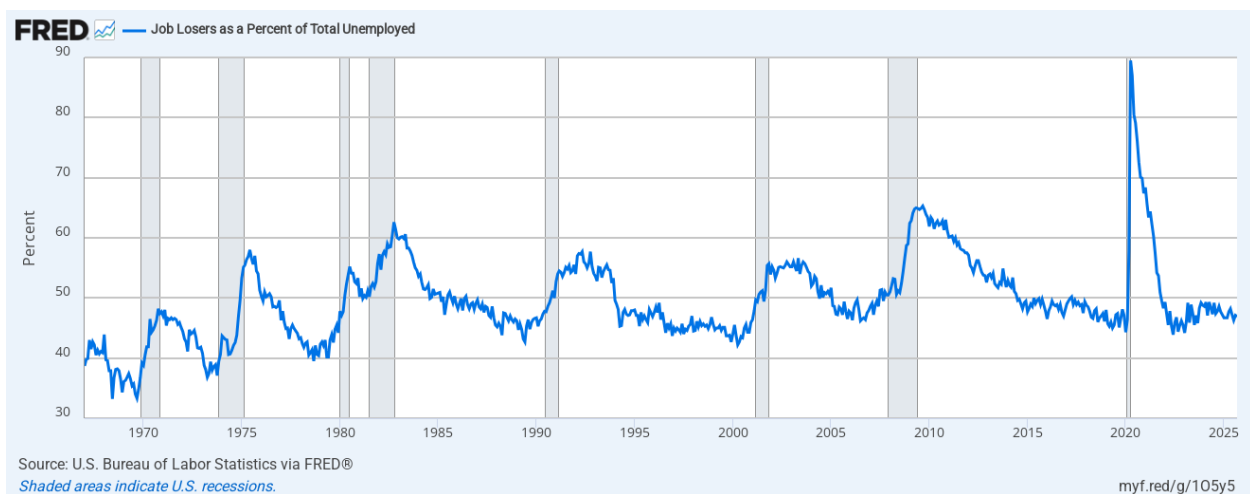


* The release dates were delayed, so we're assuming the most recent release (Sep. 2025) hasn't been observed yet

This report is on forecasting the variable job losers as a percent of total unemployed, and to interpret the economic implications of the forecast. The data is released by the U.S. Bureau of Labor Statistics as part of the monthly report of *The Employment Situation*. This variable is already seasonally adjusted and measures the proportion of people that are currently unemployed due to job loss relative to the total currently unemployed. The term job losers include layoffs, firings, and other forms of loss (U.S. Bureau of Labor Statistics, 2025). This is different from those entering the market for the first time and those who have opted to leave voluntarily, such as a new college graduate entering the market or someone quitting their job.

Looking at the time-series provided by FRED, there are a few initial observations. During recessions there is a clear jump in the rate, followed by a, typically, more gradual decline in rate. It is also apparent that the recession period around the beginning of Covid-19 saw a significantly higher jump than any other recession period. Lasting from February to April of 2020, this recession period will be denoted as the covid shock for the remainder of the report. Outside of recessions, it appears that there also might be an overall upwards trend in the data. These initial observations have influenced the steps taken to model the forecast.



The first step in determining a good forecasting model for this data is deciding whether this data exhibits stationary behavior. While there seems to be a slight upwards trend, the increase marked by each recession followed by a recovery period towards the supposed trend signifies a change in mean during the recessions. This could be a potential indication of a non-stationary mean in the time-series. Accordingly, an ADF (augmented Dickey-Fuller) test was run to test for the possibility of a unit root. The test was statistically significant, suggesting that there is no unit root in the data, and that it is stationary. However, when also plotting the ACF (autocorrelation function) and the PACF (partial autocorrelation function), the

data also exhibits highly persistent, AR(1)-like, behavior. These plots could also be potential indications of non-stationary data under the assumption that there was a unit root. While the ADF test suggested otherwise, it should be noted that the data was already seasonally adjusted, which could impact the results of the test. Thus, both directions are explored when modeling the data.

Under the assumption that there is no unit root, it was noted that there appears to be a change in trends during recessions, as opposed to outside of a recession. Due to this, it is reasonable to explore the impact of a mean shift on the (assumed stationary) model. Dummy variables were created for all the recession periods as well as an additional dummy variable for the covid shock. Performing OLS (ordinary least squares) regression on these dummy variables with respect to the percentage of job losers suggested these are both statistically significant shifts. Although after exploring the remaining structure of the data, only the mean shift for the covid shock remained in the final model. Intuitively this would make sense, since a similar pattern is seen during each recession and the covid shock is the only period with a drastically steeper jump in the curve relative to any other period. The mean shift dummy variable for the covid shock considers the extra change in the jump, leaving the remainder of the model to describe the typical behavior in and out of recession.

Since there is an apparent trend in the overall data, OLS regression was also performed on the time variable with respect to the percentage of job losers. This resulted in the time variable appearing statistically significant, but it had a near zero coefficient (approximately 0.0133067). This would imply that the time variable has little impact on the regression. Intuitively, this follows as the trend in the time-series plot, while appearing upwards, looks very gradual. In fact, the time variable also gets omitted from the final model after exploring the remaining structure, as the proposed final model suggested the time variable ended up being statistically insignificant.

Economically speaking, it is reasonable to suspect that seasonality might also have an impact on the job loss rate. Industries like agriculture and construction tend to have less productivity in the winter months. However, the data was already seasonally adjusted. Consequentially, performing OLS regression on a series of dummy variables representing the months of the year suggested they were all statistically insignificant. Furthermore, graphing the seasonal trend showed little marginal impact on the trajectory of the data. This suggests that there is no more information to be had by modeling seasonality, and so it is not included in the final model.

After exploring the trend and mean shift behavior, the remaining structure of the data can be determined by examining the autocorrelation behavior. Both the ACF and PACF were run after removing the mean shift and trend from the data. The ACF showed a gradual decline in the lag values, while the

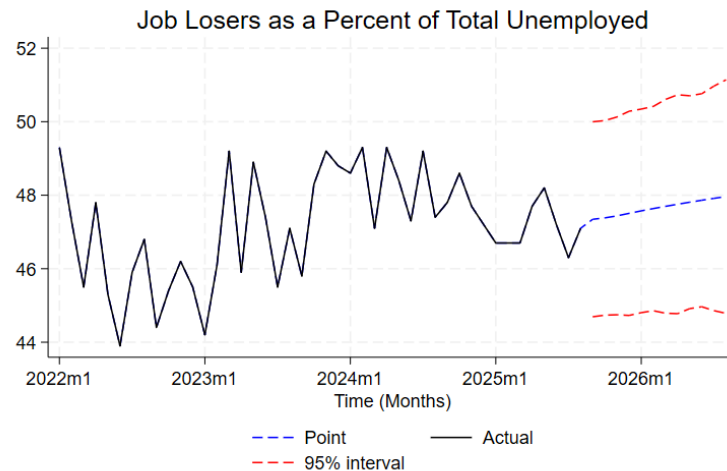
PACF showed a distinct cutoff after the first three lag values. This would imply that the remaining behavior can be modeled with an AR(3) model. With that in mind, MLE (maximum likelihood estimation) regression with respect to the percentage of job losers was performed on an AR(3) model, the time variable, and the dummy variables for the recessions and covid shock. This regression is where the time variable and recession dummy variables were statistically insignificant. After omitting the insignificant variables, all that remains is the AR(3) model with the covid shock dummy variable.

Including only the covid shock dummy variable, the AIC (Akaike information criterion) was also measured on several AR models with varied amounts of lag terms. By opting for the model with the lowest AIC, the tests also suggested an AR(3) model. On plotting the ACF and PACF of the AR(3) with the covid shock variable, the remaining structure is effectively white noise.

Under the assumption that there is no unit root, the final model that was used for forecasting is an AR(3) model with the covid shock dummy variable. The parameter values are in the Stata regression table output below. Looking at the coefficients of the lag terms, the first lag has the greatest impact. The 2nd lag term has a negative coefficient, which tends to drag the data back down after increasing. This models the incline seen during recessions followed by an inevitable decline or recovery in the data. The covid shock dummy variable accounts for the extra steep incline seen in that recession, better reflecting the cycles observed at the start of every other recession period.

job_lovers	OPG					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
job_lovers						
covid_shock	4.201794	1.49037	2.82	0.005	1.280724	7.122865
_cons	49.09188	2.048571	23.96	0.000	45.07675	53.10701
ARMA						
ar						
L1.	1.083198	.0370253	29.26	0.000	1.01063	1.155766
L2.	-.1984971	.0499708	-3.97	0.000	-.2964382	-.1005561
L3.	.0757287	.0337653	2.24	0.025	.00955	.1419074
/sigma	1.834754	.0253656	72.33	0.000	1.785039	1.88447

Using this model, the job losers as a percentage of total unemployed were iteratively forecasted for the next year using a 95% confidence interval. The forecast projects an overall upwards trend in the percentage. Particularly, for the month of September in 2025, the predicted value is approximately 47.34519. This is a little higher than the percentage for August in 2025 (the most recent data point observed), which is rounded to 47.1.



The implication of a forecasted upwards trend would suggest potential volatility in the job market. An upwards trend in job loss could indicate early signs of a lower labor demand. While there could be various causes, it opens the door to future avenues of exploration. One proposed avenue is how technological growth among companies impacts job loss and higher factors. It is also worth noting that while there are not any significant jumps in the forecast, even a small change in the percentage could have a significant impact.

There are various studies exploring job loss and unemployment risk in the economy. In recent news, there is an increase in public concern on how the growth in AI might disrupt the job market and lead to unemployment. Although there tends to be many variable factors influencing job loss, they can often get overlooked. There has been a growing trend in unemployment risk, often due to technological and automation advances (Frank, Ahn, & Moro, 2035). This can be worrisome for many. Which is why paying attention to this variable could prove important, while also providing insights into unemployment patterns as whole. This variable exhibits a correlation with the recession periods. Thus, forecast for job loss could expand beyond unemployment and provide insights about the overall economy.

The suggested correlation between recessions and job loss prompts the remainder of the forecast analysis. The first forecast was done under the assumption that there is no unit root, and that the data is stationary. Now suppose that the data was incorrectly classified by the ADF test, given the prior skepticism discussed above. By first-order differencing the job loss variable, all that remains when plotting the ACF and PACF is effectively white noise. The differenced variable also does not have a unit root by the ADF test, which contributes to the possibility that the original data may have been non-stationary in the first place. This would suggest the time-series is effectively a random walk and is integrated of order one, $I(1)$.

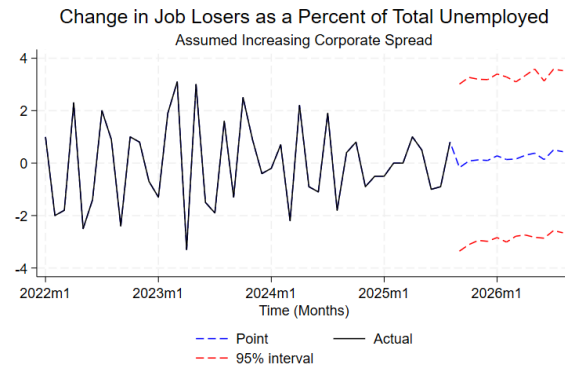
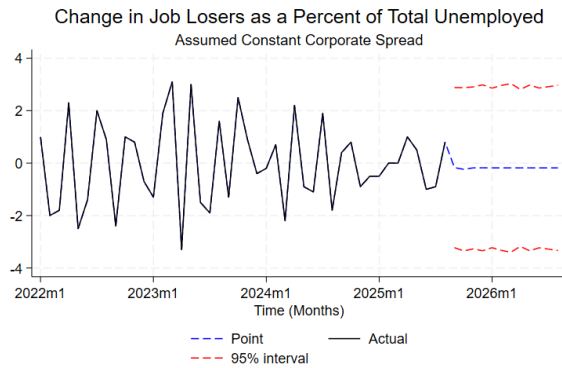
Differencing the variable provides more potential forecast models, although the interpretation of them is different from the previous forecast. This assumes that the data was non-stationary. By taking the difference, the forecast is now interpreted as the change in percentage of the variable. Thus, it is the change in percentage of job losers relative to the total unemployed. A positive rate suggests that the percentage of job losers is increasing by that rate in the given month, and vice versa. Effectively measuring the volatility in growth relative to the total unemployment rate. Consequently, the following models can't be directly compared to those from the previous forecast when selecting a model.

While the differenced variable could be modeled as a simple random walk, the apparent correlation with recessions suggests that there are leading indicators that might better predict the change in job loss. The ideal leading indicator would predict the effects of a recession. As such, the following potential indicators were evaluated: 1-year and 10-year treasury yield spread, corporate credit spread, and the change in treasury yield over a 3 and 12-month period. A series of ADL (Autoregressive Distributed Lag) models were evaluated using Grainger-Causality and the AIC. Ultimately, the results suggested that the corporate spread is the best indicator of the five that were tested.

Opting for a simpler model to avoid overfitting, the final model is an ADL(2, 2) model using the corporate spread indicator and the covid shock dummy variable. This model minimizes the BIC (Bayes Information Criterion) while also aligning with the lowest AIC up the second-order lags. The parameter values are in the Stata regression table output below.

d_job_losers	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
d_job_losers						
L1.	-.0289436	.0325823	-0.89	0.375	-.0929151	.0350279
L2.	-.0546201	.0319589	-1.71	0.088	-.1173678	.0081275
corporate						
L1.	2.544408	.5234654	4.86	0.000	1.516645	3.572171
L2.	-2.179287	.5208573	-4.18	0.000	-3.20193	-1.156645
covid_shock	14.92828	.9166881	16.29	0.000	13.12847	16.72809
_cons	-.4355438	.1621349	-2.69	0.007	-.7538766	-.1172109

Two iterative forecasts were made using this selected model with a 95% confidence interval. The first assumes that the corporate spread remains constant over the forecasted period. The second forecast assumes an increasing corporate spread over the next year. This increase is calculated at a linear increase of 0.1 plus a normally distributed random variable with mean 0 and standard deviation 0.05. An increase in corporate spread would suggest a recession is more likely, as it is implying a perceived riskier investment for corporate bonds.



Some observations can be made by looking at the forecast plots. Particularly, the forecast for the next month shows a decline under both assumptions. In both scenarios, the predicted change for September 2025 is approximately -0.1722. This would suggest that the job losers as a percent of total unemployed will decrease in the next month by about 0.17%. It can also be observed that an increase in corporate spread suggests a positive increase in the rate of job loss. Intuitively, this follows as it simulates a higher recession risk, which historically should lead to an increased rate in job loss by referencing the time-series plot provided by FRED.

Comparing this forecast to the first one, the predicted absolute change over the next month lies in a similar margin for both, but they suggest different directions in growth. Although the ADF suggested there is no unit root, the first-order differencing could be a stronger indicator of non-stationary data. Thus, while both models could be acceptable, future exploration should be taken in determining whether the true DGP (data generating process) is stationary or not.

It could also be argued that the differenced model is better at capturing the recession behavior of the true DGP. This is since it utilizes a recession indicator as opposed to relying only on past observations of the DGP. Although, this indicator can fall short when forecasting into the future. As seen in the above time-series plots, the forecast depends heavily on assumptions about the behavior of the corporate spread in the future, as the current model only has lag terms up to order 2. However, it does open the door to modeling how future recession effects could alter the forecast, which isn't well-achieved with the first forecast model.

Regardless of the chosen model, forecasting job losers as a percent of total unemployed provides insights into unemployment risk and broader economic conditions. The analysis shows that while both the stationary and non-stationary models can offer meaningful perspectives, more testing will need to be done in future forecast analyses. Both forecast models suggest modest changes in the near term, but potential for increased volatility suggests the importance of watching this variable, especially as technological

advancements and change in corporate risk perceptions shape trends. Ongoing evaluation of this variable and related economic indicators will be crucial for anticipating future labor market developments and policy making.

References

- Frank, M. R., Ahn, Y.-Y., & Moro, S. (2035, April 2). AI exposure predicts unemployment risk: A new approach to technology-driven job loss. (E. Kimbrough, Ed.) *PNAS Nexus*, 4(4). doi:10.1093/pnasnexus/pgaf107
- U.S. Bureau of Labor Statistics. (2025). *The Employment Situation — September 2025*. Retrieved from <https://www.bls.gov/news.release/pdf/empst.pdf>
- U.S. Bureau of Labor Statistics. (n.d.). *Job Losers as a Percent of Total Unemployed [LNS13023622]*. FRED, Federal Reserve Bank of St. Louis. Retrieved from <https://fred.stlouisfed.org/series/LNS13023622>

Stata Code

```
*****
* Data Links:
* https://fred.stlouisfed.org/series/LNS13023622
* https://fred.stlouisfed.org/series/USRECM
* https://fred.stlouisfed.org/series/AAA
* https://fred.stlouisfed.org/series/BAA
* https://fred.stlouisfed.org/series/GS1
* https://fred.stlouisfed.org/series/GS10
* https://fred.stlouisfed.org/series/TB3MS

* Input files: LNS13023622.csv, USRECM.csv, AAA.csv, BAA.csv, GS1.csv, GS10.csv, TB3MS.csv

* Output files: forecast.png, forecast_constant, forecast_analysis.png
* job_losers_sa.dta, USRECM.dta, AAA.dta, BAA.dta, GS1.dta, GS10.dta, TB3MS.dta
*****

clear all
set more off
capture log close
global proj_dir "proj-dir-path"
log using "${proj_dir}\forecast_analysis.log", replace text

* Setup
import delimited "${proj_dir}\LNS13023622.csv", clear
rename lns13023622 job_losers_sa
save "job_losers_sa.dta", replace
import delimited "${proj_dir}\USRECM.csv", clear
save "USRECM.dta", replace
import delimited "${proj_dir}\AAA.csv", clear
save "AAA.dta", replace
import delimited "${proj_dir}\BAA.csv", clear
save "BAA.dta", replace
import delimited "${proj_dir}\GS1.csv", clear
save "GS1.dta", replace
import delimited "${proj_dir}\GS10.csv", clear
save "GS10.dta", replace
import delimited "${proj_dir}\TB3MS.csv", clear
save "TB3MS.dta", replace

use job_losers_sa.dta, clear
local files : dir . files "*.dta"
foreach file in `files' {
    merge 1:1 observation_date using `file', nogen
}
drop if missing(job_losers_sa)

gen t = mofd(date(observation_date, "YMD"))
format t %tm
tsset t
drop if t >= ym(2025, 9)
```

```

gen covid_shock = (t >= tm(2020m2) & t <= tm(2020m4))
gen month = month(dofm(t))

gen spread1 = gs1 - tb3ms
gen spread2 = gs10 -tb3ms
gen corporate = baa - aaa
gen dt3 = tb3ms - L.tb3ms
gen dt12 = gs1 - L.gs1

*-----
describe
list in 1/10
tsline job_losers_sa usrecm
*-----

/*
  Modeling
  -----*/
* Denote: SS = Statistically Significant,
* SI = Statistically Insignificant

* Stationarity
dfuller job_losers_sa, regress // SS, _cons SI
ac job_losers_sa // Slow Decay
pac job_losers_sa // Cutoff after lag 1
// Conflicting results could be due to SA

** No Unit Root Assumption
* Mean Shift
reg job_losers_sa covid_shock // SS
predict mean_shift, xb
predict demeaned_sa, residual
* Trend
reg demeaned_sa t // SS but t coef 0.0133067
predict trend, xb
tsline trend job_losers demeaned_sa
predict demeaned_detrended_sa, residual
* Remaining Seasonality
reg demeaned_detrended i.month // SI
predict seasonal, xb
tsline seasonal // Marginal differences
* Stationarity || Demeaned & Detrended
dfuller demeaned_detrended_sa, regress // SS, _cons SI
ac demeaned_detrended_sa // Slow Decay
pac demeaned_detrended // Cutoff after 3
// Suggests AR(3)
arima demeaned_detrended_sa, arima(3,0,0)
predict resid_ar3, residual
ac resid_ar3 // White Noise
pac resid_ar3 // White Noise

* AR(3) + Trend + Recession/Covid Dummies
arima job_losers usrecm covid_shock t, arima(3,0,0) // t, usrecm are SI

```

```

* AR(3) + Trend + Covid Dummy
arma job_losers covid_shock t, arima(3,0,0) // t is SI
* AR(P) + Covid Dummy
forvalues i = 1/3 {
    arma job_losers covid_shock, arima('i',0,0)
    estimates store AR'i'_model
}
arma job_losers covid_shock, arima(3,0,0)
predict resid_ar3_cov, residual
ac resid_ar3_cov // White Noise
pac resid_ar3_cov // White Noise

** Existing Unit Root Assumption
* Stationarity || Differencing
dfuller D.job_losers, regress // SS, _cons SI
ac D.job_losers // White Noise
pac D.job_losers // White Noise
// Suggests Random Walk, I(1)

* Differencing Models
arma D.job_losers_sa
estimates store diff_model
arma D.job_losers_sa, noconstant
estimates store diff_model_no_cons

** Leading Indicators
dfuller spread1 // SS
dfuller spread2 // SS
dfuller corporate // SS
dfuller dt3 // SS
dfuller dt12 // SS

local indicators dt3 dt12 spread1 spread2 corporate

forvalues l = 1/20 {
    foreach var of local indicators {
        // Granger Causality
        quietly regress D.job_losers_sa L(1/'l').D.job_losers_sa L(1/'l').'var'
        estimates store m_'var'_'l'
        testparm L(1/'l').'var'

        // Models with Covid Mean Shift
        quietly regress D.job_losers_sa L(1/'l').D.job_losers_sa L(1/'l').'var' covid_shock
        estimates store m_'var'_'l'_cov
        testparm L(1/'l').'var' covid_shock
    }
}

/*
Model Selection
-----*/

```

```

estimates stats _all

** No Unit Root Assumption
arima job_losers_sa covid_shock, arima(3,0,0)
estimates store Final_Model_No_Unit_Root
estimates stats

** Unit Root Assumption // Differencing
regress D.job_losers_sa L(1/3).D.job_losers_sa L(1/3).corporate covid_shock
// Keeping only SS corp lag coefs
gen d_job_losers = D.job_losers_sa
regress d_job_losers L(1/2).d_job_losers L(1/2).corporate covid_shock
predict res, residual
pac res
estimates store Final_Model_Unit_Root
estimates stats

/*
  Forecasting
-----*/

set seed 2004
tsappend, add(12)

replace covid_shock = 0 if missing(covid_shock)

forecast create model, replace
forecast estimates Final_Model_No_Unit_Root
forecast solve, simulate ( errors betas , statistic (stddev , prefix (sd_)) reps(1000))

gen L_int = f_job_losers_sa + invnormal(0.025)*sd_job_losers_sa
gen U_int = f_job_losers_sa + invnormal(0.975)*sd_job_losers_sa

tsline f_job_losers_sa job_losers_sa L_int U_int if t >= ym(2022, 1), legend(order(1 2 3) ///
label(1 "Point") label(2 "Actual") label(3 "95% interval") ) ///
lcolor(blue black red red) lpattern(dash solid dash dash) ///
xtitle("Time (Months)") title("Job Losers as a Percent of Total Unemployed")
graph export forecast.png, replace

capture drop f_* sd_* L_int* U_int*
// Assumption, no change
drop if _n > 704
tsappend, add(12)

replace covid_shock = 0 if missing(covid_shock)
replace corporate = L.corporate if missing(corporate)

forecast create model, replace
forecast estimates Final_Model_Unit_Root
forecast solve, simulate ( errors betas , statistic (stddev , prefix (sd_)) reps(1000))

gen L_int = f_d_job_losers + invnormal(0.025)*sd_d_job_losers

```

```

gen U_int = f_d_job_losers + invnormal(0.975)*sd_d_job_losers

tsline f_d_job_losers d_job_losers L_int U_int if t >= ym(2022, 1), legend(order(1 2 3) ///
label(1 "Point") label(2 "Actual") label(3 "95% interval") ) ///
lcolor(blue black red red) lpattern(dash solid dash dash) ///
xtitle("Time (Months)") title("Change in Job Losers as a Percent of Total Unemployed") ///
subtitle("Assumed Constant Corporate Spread")
graph export forecast_constant.png, replace

capture drop f_* sd_* L_int* U_int*
// Assumption, increasing corporate spread
drop if _n > 704
tsappend, add(12)
replace covid_shock = 0 if missing(covid_shock)
//replace corporate = L.corporate + 0.5 if missing(corporate)
forvalues i = 1/12 {
    replace corporate = L.corporate + 0.1 + rnormal(0, 0.05) if missing(corporate)
}

forecast create recession_model, replace
forecast estimates Final_Model_Unit_Root
forecast solve, simulate ( errors betas , statistic (stddev , prefix (sd_)) reps(1000))

gen L_int = f_d_job_losers + invnormal(0.025)*sd_d_job_losers
gen U_int = f_d_job_losers + invnormal(0.975)*sd_d_job_losers

tsline f_d_job_losers d_job_losers L_int U_int if t >= ym(2022, 1), legend(order(1 2 3) ///
label(1 "Point") label(2 "Actual") label(3 "95% interval") ) ///
lcolor(blue black red red) lpattern(dash solid dash dash) ///
xtitle("Time (Months)") title("Change in Job Losers as a Percent of Total Unemployed") ///
subtitle("Assumed Increasing Corporate Spread")
graph export forecast_recession.png, replace

```